Utrecht University

An overview of photo-current spectroscopy v1.0.0

To catch up Ioannis

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Nanophotonics

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3 Electrical Detection

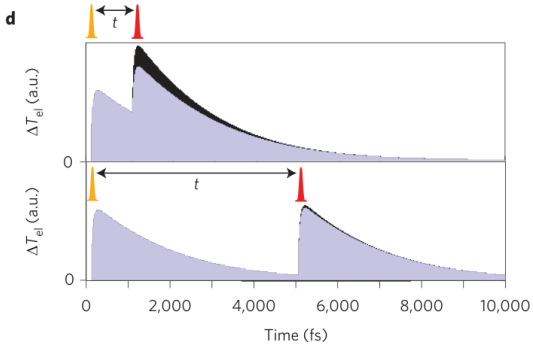
Getting Started

The principle of the measurement

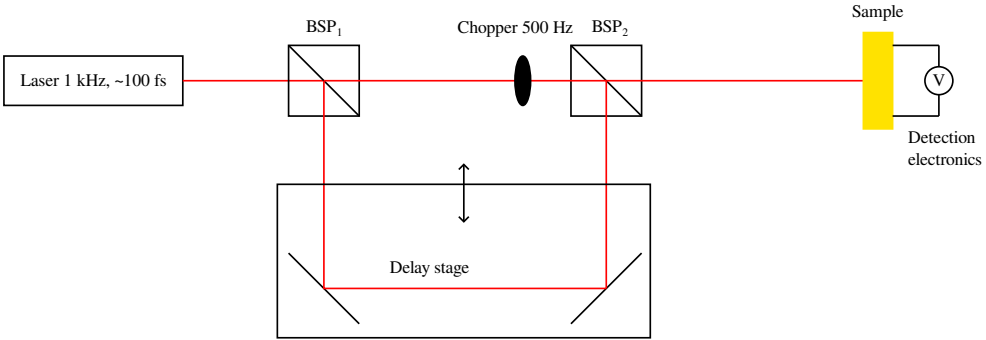
$$C_{\mathbf{e}} = \int_0^\infty d\varepsilon\, \varepsilon g(\varepsilon) \frac{\partial}{\partial T_{\mathbf{e}}} f(\varepsilon, T_{\mathbf{e}})$$

$$\Delta T_{\mathbf{tot}} = \Delta T_1 + \Delta T_2 + C$$

$$\Delta\sigma = \Delta\mu\Delta N$$



Optical setup

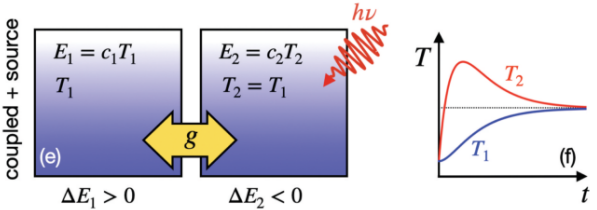


Two Temperature Model

Two temperature model

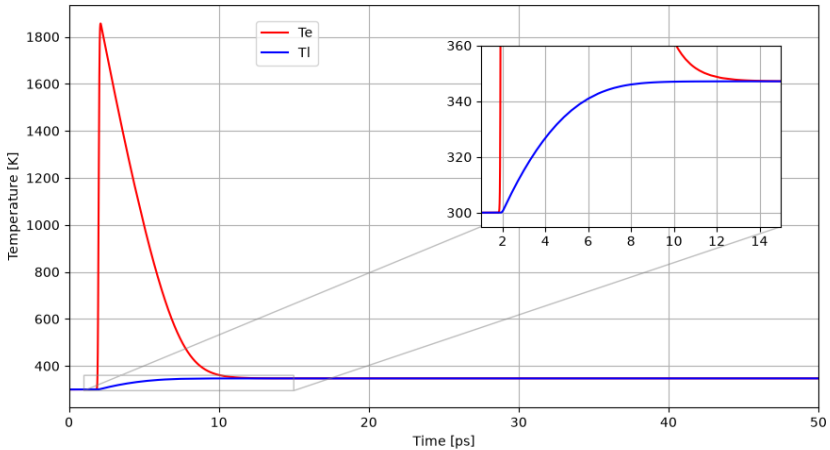
$$C_{\mathrm{e}}(T_{\mathrm{ph}})\frac{\partial T_{\mathrm{e}}}{\partial t} = g(T_{\mathrm{e}})\left(T_{\mathrm{ph}} - T_{\mathrm{e}}\right) + S(t)$$

$$C_{\mathrm{ph}}(T_{\mathrm{ph}})\frac{\partial T_{\mathrm{ph}}}{\partial t} = g(T_{\mathrm{ph}})\left(T_{\mathrm{e}} - T_{\mathrm{ph}}\right)$$



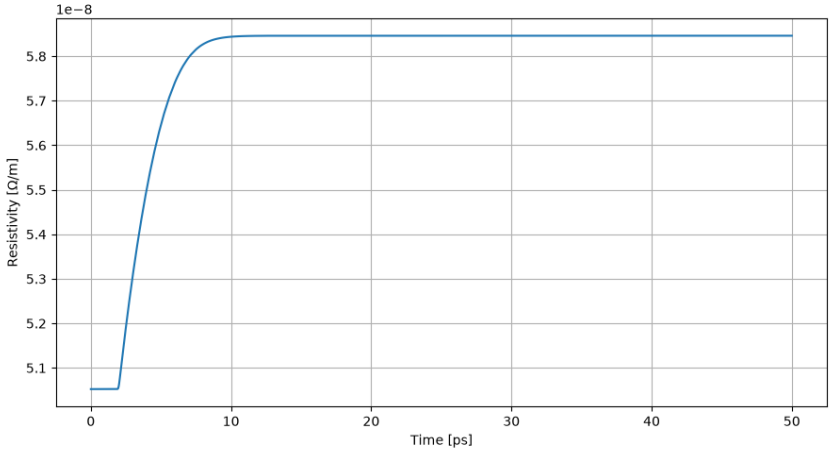
TTM results

Results under a fluence of $178.254\,\mu J\,cm^{-2}$



Resistivity

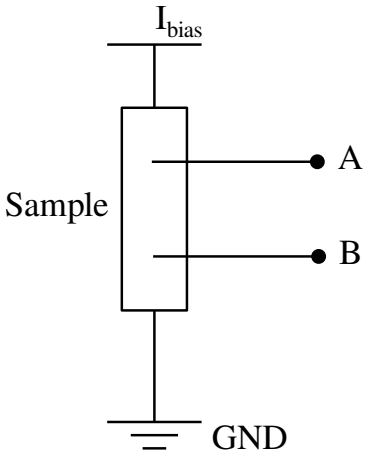
Scattering rate $\nu_{\text{tot}} = AT_{\text{ph}}$, with $A = 0.084/300\,\text{fs}^{-1}\,\text{K}$



Electrical Detection

4-point measurement

$$V_{\text{A-B}} = I_{\text{bias}} d \Delta \rho$$
$$= (100\,\text{mA}) \cdot (1\,\text{mm}) \cdot (80\,\text{n}\Omega) = 8\,\text{pV}$$



Direct current measurement

$$I = \frac{V_{\text{bias}}}{R_c + R_{\text{sample}}}$$

now with $R_{\text{sample}} << R_c$,

$$I \approx V_{\text{bias}}/R_c$$

